

File permissions in Linux

Project description

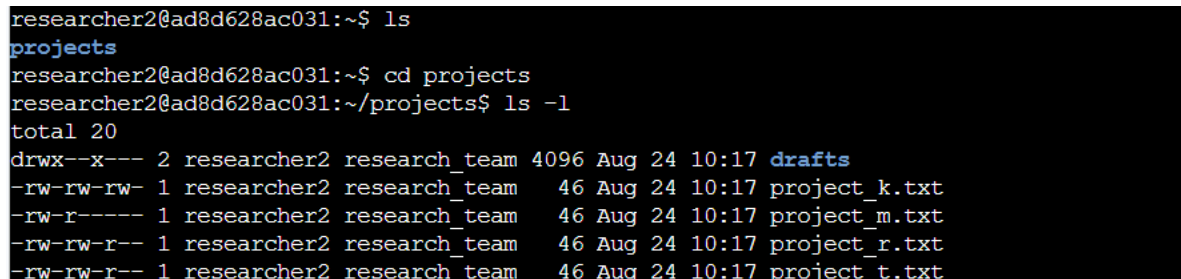
This portfolio project is to showcase my understanding and uses of Linux commands. The overview tells me that I am a security analyst assigned to manage permissions of the research team to ensure members of the teams have the correct permission at all times.

Check file and directory details

Describe the permissions string

I will describe what the permission string is and provide an image of how it works and break it down for easier understanding.

For this example I will use the `ls -l` command to view the permission string for each file within the directory:



```
researcher2@ad8d628ac031:~$ ls
projects
researcher2@ad8d628ac031:~$ cd projects
researcher2@ad8d628ac031:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Aug 24 10:17 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Aug 24 10:17 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Aug 24 10:17 project_m.txt
-rw-rw-r--  1 researcher2 research_team  46 Aug 24 10:17 project_r.txt
-rw-rw-r--  1 researcher2 research_team  46 Aug 24 10:17 project_t.txt
```

d/rwx/r-x/r-x

The D stands for directory, this stands for the directory where files are located vs the other permission strings that do not have the D at the start show that they are files located within the directory.

The second group of letters in the permission for the user 'rwx' meaning the user has read, write and execute permissions within the directory. The third and forth which are identical in the example are the same 'r-x' meaning they have read and execute permissions, they do not have write permissions meaning identifiable with the '-' in the permission string.

Change file permissions

In this part of the project I have to show my understanding of how to change file permissions, file permissions are essential for security to ensure no unauthorized activity is performed on any files within the FHS. For this I will need to use the 'chmod' command to modify permissions.

The project says that no members of the other group should have write permissions and I need to modify the permissions to correct this.

```
researcher2@ad8d628ac031:~$ ls
projects
researcher2@ad8d628ac031:~$ cd projects
researcher2@ad8d628ac031:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Aug 24 10:17 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Aug 24 10:17 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Aug 24 10:17 project_m.txt
-rw-rw-r--  1 researcher2 research_team  46 Aug 24 10:17 project_r.txt
-rw-rw-r--  1 researcher2 research_team  46 Aug 24 10:17 project_t.txt
```

Looking again at the output from the terminal we can see that the only file that the group others have write permissions is on the file named 'project_k.txt', we need to use commands to remove this permission on that file.

The commands we will need to use will be 'chmod' along with 'o-w'. The second command targets the others group using 'o' then we use '-' this is the operator needed to remove a permission for example, '+' would add the permission, and lastly we use 'w' so the command knows to remove the write permissions. Here is what the output should look like.

```
researcher2@ad8d628ac031:~/projects$ chmod o-w project_k.txt
researcher2@ad8d628ac031:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Aug 24 10:17 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Aug 24 10:17 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Aug 24 10:17 project_m.txt
-rw-rw-r--  1 researcher2 research_team  46 Aug 24 10:17 project_r.txt
-rw-rw-r--  1 researcher2 research_team  46 Aug 24 10:17 project_t.txt
researcher2@ad8d628ac031:~/projects$
```

You will now be able to see that the write permission is no longer there for others on the project_k.txt. Using 'ls -l' command again ensures that we have successfully removed the permission and have completed our task.

Change file permissions on a hidden file

Similar to before we need to change permissions to files that are hidden. In this example the file has been archived and nobody should have write permissions, but user and group should still be able to read the file. To show the hidden files in the terminal we need to use the 'ls -la' command, this will display hidden files, files starting with a period.

```
researcher2@b7da6b4f3e7d:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Aug 24 11:23 .
drwxr-xr-x 3 researcher2 research_team 4096 Aug 24 11:34 ..
-rw--w---- 1 researcher2 research_team  46 Aug 24 11:23 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Aug 24 11:23 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Aug 24 11:23 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Aug 24 11:23 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Aug 24 11:23 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Aug 24 11:23 project_t.txt
```

As we see from the output, the permissions for the user is read and write and for group is only write. The project said that only user and group should be able to read the mentioned file which is named .project_x.txt. This means we will need to remove the permission write from user, then in the group permissions we will need to remove the write permission and add the read permission. Others will remain untouched as the permission where both not mentioned and there is no permissions granted for others.

```
researcher2@b7da6b4f3e7d:~/projects$ chmod u-w,g+r,g-w .project_x.txt
researcher2@b7da6b4f3e7d:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Aug 24 11:23 .
drwxr-xr-x 3 researcher2 research_team 4096 Aug 24 11:34 ..
-r--r----- 1 researcher2 research_team  46 Aug 24 11:23 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Aug 24 11:23 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Aug 24 11:23 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Aug 24 11:23 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Aug 24 11:23 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Aug 24 11:23 project_t.txt
researcher2@b7da6b4f3e7d:~/projects$
```

Chmod – change permissions command

u-w – remove write permission for user

g+r – adds read permissions for group

g-w – removes write permission for group

A comma is used so you can perform multiple commands in one prompt.

After I used the correct command as shown in the image I used 'ls -la' to display all files again and ensure that the files now have the correct permissions.

Change directory permissions

This is similar again to the above examples only this time we target the 'drafts' directory. The project mentions that Researcher2 should be the only authorized user.

To do this we will need to select the drafts directory by using 'cd drafts'. When there we will use 'ls -la' to display all contents of drafts.

```
researcher2@b7da6b4f3e7d:~/projects$ cd drafts
researcher2@b7da6b4f3e7d:~/projects/drafts$ ls -la
total 8
drwx--x--- 2 researcher2 research_team 4096 Aug 24 11:23 .
drwxr-xr-x 3 researcher2 research_team 4096 Aug 24 11:23 ..
researcher2@b7da6b4f3e7d:~/projects/drafts$ cd ..
researcher2@b7da6b4f3e7d:~/projects$ chmod g-r,g-x,o-r,o-x drafts
researcher2@b7da6b4f3e7d:~/projects$ cd drafts
researcher2@b7da6b4f3e7d:~/projects/drafts$ ls -la
total 8
drwx----- 2 researcher2 research_team 4096 Aug 24 11:23 .
drwxr-xr-x 3 researcher2 research_team 4096 Aug 24 11:23 ..
researcher2@b7da6b4f3e7d:~/projects/drafts$
```

As seen in the image the permissions are drw – we will run the command 'chmod g-r,g-x,o-r,o-x drafts' to remove the permissions, but first we will need to back one level to the projects directory so we can preform this command correctly.

Now as you can see we navigated back to the drafts directory and ran the ls -la command again and we can now see that all permissions have been removed for everyone except the user as the project requested.

Note: I had preformed the correct command previously when doing the lab first and the commands had been removed before the screenshot, before the screenshot the permissions string read as 'drwxr-xr-x' this is the reason for the mentioned command above.

Summary

In this project I reviewed and managed permissions for the research team and the use Researcher2 by using the learned Linux commands. I used 'ls -l' and 'ls -la' to check permissions and identify hidden files and to also read and manage permissions displayed in the permission strings for users, groups and others. I have also used the 'chmod' command to remove and add permissions to ensure safety and access of information for authorized users. This ensure safety and correct level of access and helps reduce the risk of unauthorized access to files.